# KISAN-V: INDIGENOUS EDGE-AI PLATFORM

Real-Time 'See & Spray' Precision Agriculture Retrofit for Drones

## PRODUCT & INVESTMENT BRIEF EDITION

A comprehensive technical, financial, and market evaluation of KISAN-V — an indigenous, RISC-V based edge-AI retrofit platform that upgrades existing agricultural drones with real-time target recognition and micro-precision pesticide spraying capability.

### KEY PRODUCT & FINANCIAL SNAPSHOT

- **Target Chemical Savings:** 60% chemical reduction compared to blanket spraying.
- **Hardware Selling Price:** Rs. 50,000 | **Est. Manufacturing Cost (BOM):** Rs. 35,000 | **Gross Profit:** Rs. 15,000 (30% Gross Margin).
- **Implied Customer Savings:** Rs. 7.2 Lakh / year per drone (1,200 acres/yr @ Rs. 600/acre savings).
- **Payback Period:** < 1 Season (< 3 months of operational spraying).
- **Architecture:** Fully on-device Edge-AI on RISC-V processor (Zero cloud dependency)

# EXECUTIVE BRIEFING & TABLE OF CONTENTS

## Transforming Indian Agriculture Through Edge Intelligence

Indian agriculture stands at a transformative inflection point. While agricultural drones have expanded rapidly across rural districts, first-generation spray drones suffer from a fundamental flaw: they spray uniformly across entire fields, drenching healthy crops and bare soil alike. This blanket approach wastes expensive chemicals, degrades soil health, and limits farmer profitability. Advanced imported precision systems exist, but they are prohibitively expensive, locked into foreign cloud servers, and ill-suited for smallholder Indian farms.

KISAN-V addresses this challenge at its root. By delivering an affordable (Rs. 50,000) plug-and-play Edge-AI retrofit kit built on indigenous RISC-V hardware co-design, KISAN-V enables real-time 'See & Spray' capability on existing drone fleets. Operating entirely offline without cloud dependency, KISAN-V targets a 60% reduction in chemical volume, yielding Rs. 7.2 Lakh in annual savings per drone and paying for itself in under a single season.

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# CHAPTER 1: THE INDIAN AGRICULTURAL LANDSCAPE

## 1.1 The Blanket Spraying Crisis in Indian Farming

Pesticide spraying across India's 140 million hectares of arable land remains overwhelmingly blanket-based. Traditional manual backpack sprayers and conventional agricultural drones alike discharge agrochemicals continuously along spray paths, regardless of actual weed density or crop disease distribution.

This uniform spraying regime leads to massive chemical over-application. Field observations indicate that in typical Indian conditions, less than 30% of blanket-applied pesticides land on intended target weeds or pests. The remaining 70% falls onto bare earth, settles in soil, or drifts into local waterways.

## 1.2 Environmental, Health & Economic Toll

The consequences of untargeted chemical application are severe across ecological, health, and economic metrics:

- **Escalating Input Costs:** Indian farmers spend an average of Rs. 1,000 per acre on pesticides per spraying cycle. With multiple sprays per crop season, chemical costs represent a major drain on smallholder margins.
- **Soil & Water Degradation:** Chronic chemical accumulation reduces beneficial soil microbes and pollutes rural groundwater supplies.
- **Health Hazards:** Manual spraying exposes farm laborers to toxic chemical drift, causing acute respiratory and dermatological risks.

**Figure 1: Chemical Usage Comparison (Blanket vs. KISAN-V Selective Spraying)**

**Traditional Blanket Spraying:** 100% Chemical Volume Discharged (70% Wasted on Soil/Gaps) **KISAN-V Targeted 'See & Spray':** 40% Chemical Volume Discharged (100% Targeted on Weeds/Pests) **Net Targeted Chemical Reduction: 60% Savings** in agrochemical consumption per acre.

# PROBLEM DEEP-DIVE: CONVENTIONAL DRONE LIMITATIONS

### 1.3 Drones Without Decision-Making Intelligence

The emergence of agricultural drones in India marked a major leap in operational spraying speed, allowing coverage of 20–30 acres per day. However, standard commercial drones (10L–25L tank capacity) lack real-time perception. They follow pre-programmed GPS flight paths at constant speed while discharging liquid continuously through fixed nozzles.

While efficient in aerial speed, standard drones remain 'blind' to field conditions. They apply identical chemical doses over healthy crop foliage, bare gaps, rocks, and dense weed clusters alike, perpetuating the waste inherent in manual blanket spraying.

### 1.4 The Import Trap: Prohibitive Cost & Cloud Vulnerabilities

Advanced precision agriculture platforms developed in North America or Western Europe offer selective spraying, but present fatal drawbacks for the Indian ecosystem:

**CRITICAL FLAWS OF IMPORTED PRECISION SYSTEMS:**

- **Prohibitive Capital Cost:** Imported precision drones cost Rs. 8 Lakh to Rs. 15 Lakh+, far beyond the purchasing power of Indian service providers and FPOs.
- **Mandatory Cloud Connectivity:** Foreign systems rely on uploading imagery to cloud servers for AI processing. In rural India, where 4G/5G signals are intermittent or absent, cloud-dependent systems fail completely.
- **Data Sovereignty Risks:** Processing farm imagery on overseas servers creates severe security and data privacy concerns regarding national agricultural yield data.
- **Poor Crop Customization:** AI models trained on Western monoculture fields perform poorly on small, multi-cropped Indian holdings.

# CHAPTER 2: INTRODUCING KISAN-V

## 2.1 The KISAN-V Vision & Core Concept

KISAN-V is an indigenous, RISC-V based Edge-AI retrofit platform designed to bring real-time target recognition to standard agricultural drones. Rather than forcing farmers and service providers to replace their existing multi-lakh drone investments, KISAN-V upgrades existing hardware with an on-device perception and micro-actuation kit.

By combining compact optical sensors, custom RISC-V processing hardware, and high-speed solenoid nozzle triggers, KISAN-V enables true 'See & Spray' operation. The system inspects crops frame-by-frame during flight and triggers chemical sprays only when weeds or diseases are detected directly under the nozzle.

## 2.2 Why Retrofit? Upgrading Fleet Capital Efficiency

- **Massive Capital Protection:** Unlocks precision capabilities on existing 10L–25L drone frames without capital replacement.
- **Plug-and-Play Integration:** Seamlessly bolts onto existing spray boom structures in under 2 hours.
- **Democratizing Precision Tech:** Lowers the entry price of smart spraying from Rs. 10+ Lakh to an affordable Rs. 50,000 add-on.

## 2.3 Indigenous RISC-V Co-Design & Strategic Sovereignty

Built under the DIR-V (Digital India RISC-V) initiative vision, KISAN-V leverages open-standard RISC-V processor architecture. Custom hardware acceleration units execute quantized computer vision neural networks locally on the drone, eliminating reliance on imported proprietary chips and safeguarding Indian agricultural data sovereignty.

PRODUCT ARCHITECTURE & RETROFIT KIT

## 2.4 The Plug-and-Play KISAN-V AI Controller Kit

The KISAN-V system is delivered as a lightweight, self-contained hardware kit that installs onto any standard 10L, 16L, or 25L agricultural drone frame.

| Kit Component | Technical Specification | Function in KISAN-V System |
|---|---|---|
| Optical Sensing Unit | High-speed RGB / Multispectral camera module with anti-vibration mount | Captures high-frame-rate downward imagery of crop canopy in real time. |
| RISC-V Edge-AI Board | Custom embedded processor with integrated AI acceleration vector unit | Executes quantized neural networks locally with sub-10ms inference latency. |
| Micro-Actuation Controller | High-speed PWM solenoid driver circuit connected to fast-response nozzles | Triggers individual spray nozzles in milliseconds based on AI detection outputs. |
| Power & Enclosure | Vibration-damped carbon-fiber casing with DC-DC power regulation | Draws power from drone battery with negligible impact on flight endurance. |

# CHAPTER 3: TECHNICAL MECHANICS OF SEE & SPRAY

## 3.1 On-Device Edge Inference vs. Cloud Dependency

At typical spraying flight speeds of 3 to 5 meters per second (10–18 km/h) at low flight altitudes (1.5–2.5m above crop canopy), the time window between camera frame capture and nozzle passage over a target weed is less than 50 milliseconds. Cloud-based AI systems, which require 1,000 to 3,000 milliseconds for image transmission and server processing, are physically incapable of supporting real-time actuation.

KISAN-V solves this speed barrier through fully on-device edge inference. By executing lightweight, quantized neural networks directly on indigenous RISC-V hardware, KISAN-V achieves an end-to-end processing latency of under 10 milliseconds.

## 3.2 Real-Time Sub-10ms Perception & Micro-Actuation Pipeline

- **Image Capture & Preprocessing:** High-speed camera captures canopy frames under variable sunlight conditions.
- **Local Neural Inference:** Edge processor classifies foliage into crop, target weed, or bare soil within 6 milliseconds.
- **Spatial-Temporal Offset Calculation:** Algorithm calculates drone velocity and nozzle offset to time spray pulse perfectly.
- **Precision Actuation:** Solenoid driver fires a micro-pulse spray directly onto target weed with sub-centimeter accuracy.

SYSTEM WORKFLOW & PROCESSING PIPELINE

## 3.3 End-to-End Operational Execution Pipeline

The KISAN-V real-time processing loop executes continuously during flight. Below is the breakdown of stage latencies that ensure precision pulse spraying at high flight speeds:

| Pipeline Stage | Detailed Operational Function | Stage Latency |
|---|---|---|
| Stage 1: Optical Acquisition | High-FPS camera captures canopy frames under anti-vibration damping | 2.5 ms |
| Stage 2: On-Device Preprocessing | Frame normalization, contrast balancing & ROI segmentation | 1.5 ms |
| Stage 3: Neural Inference | Quantized MobileNet/YOLO model classifies weed vs crop | 4.0 ms |
| Stage 4: Spatial Timing Control | Velocity compensation & nozzle activation pulse calculation | 0.8 ms |
| Stage 5: Micro-Solenoid Spray | Fast solenoid nozzle opens for precise chemical pulse | 1.0 ms |
| **Total System Latency** | **Sub-10ms End-to-End Latency (Enables pinpoint spraying at 5 m/s flight speeds)** | **< 10.0 ms** |

# CHAPTER 4: KEY BENEFITS & UNIQUE SELLING POINTS

## 4.1 The 4 Pillars of KISAN-V Value

### 1. 60% Targeted Chemical Reduction

By replacing continuous blanket spraying with target-activated micro-pulses, KISAN-V reduces total chemical volume by up to 60%, drastically cutting input costs.

### 2. Unmatched ROI & Payback (<1 Season)

Priced at Rs. 50,000, the kit generates Rs. 7.2 Lakh in annual customer savings, achieving full payback in under 3 months of active spraying operations.

### 3. Universal Retrofit Capability

Compatible with standard commercial drone booms (10L, 16L, 25L), allowing existing drone operators to upgrade fleets without capital replacement.

### 4. 100% Offline & Reliable Operation

Zero dependency on 4G/5G internet or cloud infrastructure ensures uninterrupted spraying performance across remote Indian farmlands.

COMPETITIVE EVALUATION MATRIX

## 4.2 KISAN-V vs. Conventional Spray Drones vs. Imported Precision Systems

| Criteria | Traditional Spray Drone | Imported Precision System | KISAN-V Retrofit Kit |
|---|---|---|---|
| Spraying Method | Blanket Continuous | Selective Precision | Real-Time See & Spray |
| Processing Architecture | None (GPS Blind) | Cloud-Dependent AI | On-Device RISC-V Edge AI |
| Internet Dependency | Offline | Mandatory 4G/5G Cloud | 100% Offline Operation |
| Capital Cost | Rs. 4 Lakh – Rs. 6 Lakh | Rs. 8 Lakh – Rs. 15 Lakh+ | **Rs. 50,000 (Retrofit Add-on)** |
| Chemical Savings | 0% (Baseline waste) | 40% – 50% | **60% Targeted Reduction** |
| Annual Customer Savings | Rs. 0 | Rs. 4 Lakh – Rs. 5 Lakh | **Rs. 7.2 Lakh per drone** |
| Payback Period | N/A | 2 – 3 Years | **< 1 Season (< 3 Months)** |
| Maintenance & Service | Local / Standard | Complex / Imported Parts | Indigenous / Low Cost |

# CHAPTER 5: TARGET CUSTOMERS & MARKET SEGMENTS

## 5.1 Primary B2B Customer Segments

- **Agri-Drone Service Providers:** Entrepreneurs and spraying contractors operating commercial drone fleets across rural clusters.
- **Farmer Producer Organizations (FPOs) & Co-operatives:** Agricultural collectives seeking to lower input costs for member smallholder farmers.
- **Agri-Input & Agrochemical Companies:** Chemical manufacturers interested in precision application tools to promote sustainable product stewardship.

## 5.2 Agri-Tech Value Proposition Drivers

- **High Margin Service Fees:** Drone operators can charge premium precision spraying rates while lowering liquid payload weight per acre.
- **Extended Flight Autonomy:** 60% chemical savings means a 10L tank covers 2.5x more acreage per flight, drastically reducing reload downtime.

BUSINESS MODEL & GO-TO-MARKET STRATEGY

## 5.3 Go-To-Market & Ecosystem Partnerships

- **Direct Sales to Spraying Contractors:** Target active drone service entrepreneurs through rural agri-tech networks and university incubators.
- **OEM Bundling Partnerships:** Partner with Indian drone manufacturers to pre-install KISAN-V kits as a premium factory option.

## 5.4 Tri-Fold Revenue Architecture

### 1. Direct Hardware Kit Sales

One-time sale of the KISAN-V AI Controller Kit at Rs. 50,000 per unit.

### 2. Hardware-as-a-Service (HaaS) Subscriptions

Annual subscription model covering software model updates and hardware maintenance.

### 3. Custom Crop AI Model Licensing

Specialized neural model algorithms tailored for specific regional crops (sugarcane, cotton, paddy, soybean).

# CHAPTER 6: UNIT ECONOMICS & MANUFACTURING COSTS

## 6.1 Hardware Unit Economics Overview

The financial structure of the KISAN-V AI Controller Kit delivers robust gross margins while remaining highly attractive to customer buyers:

• **Hardware Selling Price:** Rs. 50,000 • **Estimated Manufacturing Cost (BOM):** Rs. 35,000 • **Gross Profit per Kit:** Rs. 15,000 • **Gross Profit Margin:** 30.0%

## 6.2 Detailed Bill of Materials (BOM) Cost Breakdown

| Component Category | Est. Prototype BOM Cost (Rs.) | Technical Description |
|---|---|---|
| Optical Sensor & Camera Module | Rs. 8,000 | Dual high-speed camera sensors with anti-vibration mount |
| RISC-V AI Processing Board | Rs. 14,000 | Custom embedded SoC board with neural vector engine |
| Micro-Solenoid Driver & Valves | Rs. 6,000 | Fast-response PWM solenoid valves & driver circuitry |
| Power Supply, Casing & Harness | Rs. 4,000 | Carbon housing & DC-DC step-down converter |
| Assembly, Calibration & Testing | Rs. 3,000 | Assembly labor, quality assurance & optical calibration |
| **Total Estimated BOM Cost** | **Rs. 35,000** | **Prototype estimation baseline** |

CUSTOMER ROI & FINANCIAL PAYBACK CALCULATIONS

## 6.3 Customer Turnover & Savings Assumptions

| Parameter Description | Reference Value | Analytical Notes & Calculation |
|---|---|---|
| Avg. Acres Sprayed / Drone / Year | 1,200 acres | Standard operational throughput for active spray drone |
| Avg. Pesticide Cost / Acre | Rs. 1,000 | Baseline chemical spend per spraying cycle |
| Target Cost Saving with KISAN-V | Rs. 600 / acre | Implied 60% chemical volume reduction |
| Annual Savings per Drone | **Rs. 7.2 Lakh** | Calculation: 1,200 acres × Rs. 600/acre savings |
| Payback Period | **< 1 Season** | Break-even reached after spraying ~83.3 acres (<15 days) |

## 6.4 Customer Payback Period & Financial Breakdown

- **Rapid Capital Amortization:** At Rs. 600/acre in chemical savings, a drone service provider earns back the Rs. 50,000 kit investment after spraying just 83.3 acres.
- **Multiplier Effect for Fleets:** An operator with 5 drones saves Rs. 36 Lakh annually in chemical inputs, creating an overwhelming commercial incentive for adoption.

# CHAPTER 7: PRODUCT ROADMAP & MILESTONES

## 7.1 Development Milestone Tracker

| Development Phase | Status | Key Achievements & Objectives |
|---|---|---|
| Concept & Feasibility | Completed | Validated market demand, problem definition & core concept. |
| System Architecture & Research | Completed | Finalized RISC-V co-design architecture & sensor selection. |
| Prototype Development | On Going | Functional bench testing of optical hardware & neural models. |
| Field Trials & Validation | On Going | Field testing in regional crop plots for algorithm calibration. |
| Pilot Deployment | Planned | Pre-commercial rollout of 5–10 kits with select FPOs. |

## ]7.2 Key Technology Validation Targets

- **Sub-10ms Inference Speed:** Ensure latency remains stable under full flight velocity.
- **90%+ Target Recognition Accuracy:** Achieve high precision on weed detection across sugarcane and soybean canopy trials.

FUNDING TRACTION & INVESTMENT PROPOSAL

## 7.3 Capital Raised to Date

| Source of Funds | Funds Invested (Rs.) | Allocation Focus |
|---|---|---|
| Self-Funded (Founders) | Rs. 1,00,000 | Initial hardware research & core components |
| TADASTITHI GmbH, München | Rs. 50,000 | Prototype building & optical testing |
| **Total Funds Invested** | **Rs. 1,50,000** | **Complete functional prototype baseline** |

## 7.4 Capital Request & Use of Funds

**SEEKING INCUBATION, GRANTS & SEED INVESTMENT UP TO Rs. 10,00,000**

**Proposed Capital Deployment Plan:**

- **Field Trials & Agricultural Validation (Rs. 5,0,000):** Multi-crop field testing with regional agricultural universities.
- **IP Filing & Patent Protection (Rs. 1,00,000):** Patent filings around hardware-software co-design & RISC-V acceleration.
- **Pilot Pre-Commercial Batch (Rs. 3,50,000):** Building 10 pre-commercial pilot kits for active field testing with local FPOs.

# CHAPTER 8: LEADERSHIP TEAM & GOVERNANCE

## 8.1 Co-Founders & Key Competencies

**Ashish Suryawanshi — Co-Founder | Strategy & Business Lead**

• Marketing strategy & go-to-market execution planning.
• Budgeting, cost structure definition & financial forecasting.
• Stakeholder coordination across agricultural universities and incubation centres.
• Product vision & ground-level problem validation.

**Pratik Suryawanshi — Co-Founder | Technical Lead**

• Embedded systems architecture & RISC-V co-design.
• Computer vision algorithm development & neural network quantization.
• Micro-actuation hardware integration & solenoid driver control.
• Prototype assembly, optical sensor calibration & field testing.

# CHAPTER 9: TECHNICAL & BUSINESS GLOSSARY

## Part 1: Essential Business Concepts Explained

**1. USP (Unique Selling Proposition)**
**Definition:** The distinct feature or benefit that sets a product apart from all market competitors.
**KISAN-V Example:** Fully on-device real-time See & Spray retrofit capability at Rs. 50,000 with 60% chemical savings.
**Why It Matters:** Gives drone operators an unbeatable economic advantage over legacy blanket sprayers.

**2. BOM (Bill of Materials)**
**Definition:** An exhaustive list of raw materials, components, assemblies, and sub-assemblies required to manufacture a product.
**KISAN-V Example:** The Rs. 35,000 estimated prototype manufacturing cost breakdown covering cameras, RISC-V board, solenoids, and casing.
**Why It Matters:** Directly dictates manufacturing gross margin and hardware scaling profitability.

**3. Gross Margin**
**Definition:** The percentage of total sales revenue that a company retains after incurring direct costs of production.
**KISAN-V Example:** Selling price Rs. 50,000 minus BOM cost Rs. 35,000 yields Rs. 15,000 gross profit (30.0% Gross Margin).
**Why It Matters:** High gross margins provide necessary cash flow for continuous R&D; and field support.

**4. Payback Period**
**Definition:** The length of time required for an investment to generate cash flows sufficient to recover its initial cost.
**KISAN-V Example:** Customer recovers the Rs. 50,000 kit cost in under 1 season after spraying just 83.3 acres (saving Rs. 600/acre).
**Why It Matters:** Eliminates financial adoption friction for cost-sensitive drone service providers.

**5. HaaS (Hardware-as-a-Service)**
**Definition:** A subscription model where customers pay recurring fees for hardware, software updates, and maintenance rather than purchasing outright.
**KISAN-V Example:** Optional annual maintenance and crop-specific AI model updates provided to drone operators.
**Why It Matters:** Generates predictable recurring revenue beyond one-time hardware sales.

**6. RISC-V**
**Definition:** An open-standard instruction set architecture (ISA) based on established reduced instruction set computer principles.
**KISAN-V Example:** The custom processing chip in the KISAN-V kit uses an indigenous RISC-V architecture vector engine.
**Why It Matters:** Delivers high energy efficiency, custom co-design capability, and total sovereignty from foreign silicon vendors.

**7. Edge AI**
**Definition:** The deployment of artificial intelligence algorithms directly on local hardware devices without requiring cloud connectivity.
**KISAN-V Example:** Image classification and weed detection run directly on the drone controller board during flight.
**Why It Matters:** Guarantees sub-10ms processing speeds and 100% operational reliability in rural offline fields.

**8. Computer Vision (CV)**
**Definition:** A field of AI that enables computers and systems to derive meaningful information from digital images or video.
**KISAN-V Example:** Quantized neural network models inspecting canopy frames at high FPS to identify weeds vs crops.
**Why It Matters:** Replaces human visual inspection with automated, sub-centimeter targeted detection.

**9. Retrofit**
**Definition:** The addition of new technology or features to older or existing systems without replacing the base platform.
**KISAN-V Example:** KISAN-V bolts onto standard 10L–25L drone booms in under 2 hours without replacing the drone frame.
**Why It Matters:** Protects existing capital investments and drastically lowers adoption cost.

**10. On-Device Inference**
**Definition:** The execution of trained AI neural network models directly on embedded hardware at the edge.
**KISAN-V Example:** Neural calculations executed locally within 4ms to trigger micro-solenoid valves in real time.
**Why It Matters:** Eliminates latency delays and security vulnerabilities associated with remote cloud processing.

# CHAPTER 10: RISK ANALYSIS & LIMITATIONS

## 10.1 Rigorous Risk Assessment & Mitigation Strategies

| Risk Category | Specific Risk Description | Mitigation Strategy & Status |
|---|---|---|
| Technical & Sensor | Lens dust/drift fouling optical camera vision during low flight. | Air-curtain protective shielding & automated lens check diagnostic alert. |
| Agronomic AI | Diverse weed species causing false positives in dense canopies. | Continuous dataset expansion & ongoing multi-crop university field trials. |
| Environmental | Extreme wind or low light impairing optical tracking accuracy. | Operational flight envelope guidelines limiting spraying to optimal conditions. |
| Market Adoption | Resistance from service providers accustomed to uniform acreage fees. | Demonstrating higher gross margins per hour and premium precision pricing. |

## 10.2 Ongoing Field Validation Requirements

- **Agronomic University Collaboration:** Conducting structured trial plots with agricultural scientists to quantify exact chemical reduction per crop type.
- **Environmental Durability:** Testing optical sensors against extreme dust, humidity, and operational vibrations encountered in Indian farming regions.

# INVESTOR & PARTNER CALL TO ACTION

## 11.1 Why Partner with KISAN-V

- **Grounded Technology:** Tackling a massive, urgent problem in Indian agriculture with real-time AI and indigenous RISC-V hardware.
- **Strategic Positioning:** Positioned at the intersection of AgriTech, Edge AI, and National Food Security initiatives.

## 11.2 Partnership & Investment Framework

**ENGAGEMENT OPPORTUNITIES FOR INVESTORS & INCUBATORS:**

1. **Grant & Seed Capital Allocation:** Participate in initial seed funding for field validation and pilot kit production.
2. **Incubation & Mentorship:** Provide strategic guidance, scaling support, and connections to national agricultural networks.
3. **Strategic Joint Ventures:** Collaborate with drone OEMs and service networks to integrate KISAN-V kits into active commercial fleets.

**PROJECT LEADERSHIP & CONTACT DETAILS:**

**Ashish Suryawanshi** — Co-Founder | Strategy & Business Lead
**Pratik Suryawanshi** — Co-Founder | Technical Lead

**Project Name:** KISAN-V: Indigenous Edge-AI Precision Agriculture Platform

*"Transforming Indian Agriculture, One Precise Drop at a Time."*